

Project Proposal

PowerGuard: AI-Based Energy Theft Detection

AI-Powered Electricity Anomaly Detection & Predictive Intelligence Platform

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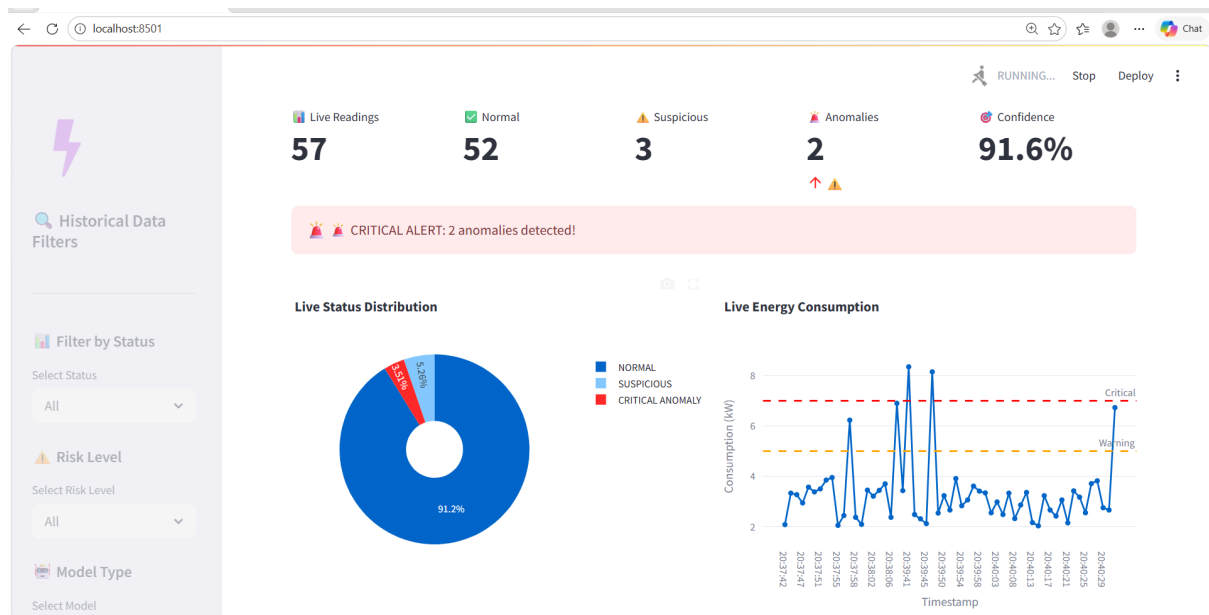
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1)Project Overview

PowerGuard is a unique initiative that concentrates on the development of an innovative system that can automatically detect and prevent energy theft through the utilization of artificial intelligence. Energy theft is a frequent phenomenon, which results in severe financial implications for utility companies. The system is expected to use sophisticated machine learning techniques such as Neural Network (NN) and SMOTE in order to be able to recognize any anomalies in energy usage and thus detect any cases of unauthorized energy usage.

In terms of energy theft detection techniques used today, the process is time-consuming and involves many errors. PowerGuard will significantly improve these aspects by ensuring fast and automatic detection of energy theft incidents. Moreover, the system will offer a Streamlit dashboard for its users.



2)Project Objective

The purpose of PowerGuard is to create an AI system that can detect cases of electricity theft through real-time monitoring of electricity usage. This involves:

1. Detection of electricity theft based on anomalies in electricity usage patterns.
2. Classification of electricity usage patterns as either normal or anomalous (electricity theft).
3. Real-time alerts and insights via Streamlit dashboard for informed decision-making.

3)Methodology:

In order to reach the goals specified above, PowerGuard will go through the following stages:

- **Data Collection & Preprocessing and Feature Engineering:**
The dataset is generated from smart meter simulations and includes features such as electricity consumption, temperature, humidity, wind speed, and previous consumption. The data contains no missing or duplicate values, but it is imbalanced because normal consumption cases exceed anomalous ones. This issue will be handled using the SMOTE technique. Feature engineering is also applied by creating a new feature called **Consumption Difference**, which represents the change between current and previous electricity consumption values.
- **Model Training:**
Two approaches will be used for the training: neural networks (NN) and Random Forests to classify cases on whether they are anomalous (energy theft) or not. Also, LSTM (Long Short-Term Memory) networks will be used for time-series predictions of energy consumption.
- **Smart Meter Real Time Simulation**

The PowerGuard system uses simulated smart meter data to mimic real-time electricity consumption, including normal usage and abnormal spikes representing possible energy theft. Environmental factors such as temperature and humidity are also considered to create realistic conditions.
- **Expert System :**

The data is analyzed using Random Forest and LSTM models for anomaly detection, while an Expert System combines their predictions to generate accurate decisions and risk levels.
- **Real-time dashboard:**

The data can be visualized in a real-time Streamlit dashboard to detect anomalies, which could then lead to quick action by the operators.

